

COPYRIGHT RESERVED

**BCA (Sem-2) – Program-
ming in Java (201) Core-III**

2024

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section-A

Compulsory

1. Choose the correct alternative for each of the following questions : $2 \times 10 = 20$

(i) Which statement is true about Java ?

- a) Java is a sequence-dependent programming language

S309/9/7

(1)

(Turn over)

- b) Java is a code dependent programming language
- c) Java is a platform-dependent programming language
- ☒ d) Java is a platform-independent programming language

(ii) Which component is used to compile, debug and execute the java programs?

- a) JRE
- b) JIT
- ☒ c) JDK
- d) JVM

(iii) Which one of the following is not a Java feature?

- a) Object-oriented
- ☒ b) Use of pointers

- c) Portable
 - d) Dynamic and Extensible
- (iv) Which environment variable is used to set the java path?
- a) MAVEN-Path
 - b) JavaPATH
 - c) JAVA
 - ☒ d) JAVA_HOME
- (v) What is not the use of "this" keyword in Java?
- a) Referring to the instance variable when a local variable has the same name
 - ☒ b) Passing itself to the method of the same class
 - c) Passing itself to another method
 - d) Calling another constructor in constructor chaining

(vi) Which of the following is a type of polymorphism in Java Programming?

- a) Multiple polymorphism
- ☒ b) Compile time polymorphism
- c) Multilevel polymorphism
- d) Execution time polymorphism

(vii) Which exception is thrown when java is out of memory?

- a) Memory Error
- ☒ b) Out Of Memory Error
- c) Memory Out Of Bounds Exception
- d) Memory Full Exception

(viii) Which of the following is a superclass of every class in Java?

- a) Array List
- b) Abstract class

S309/9/7

(4)

Contd.

- ☒ c) Object class
- d) Strin

(ix) Which one of the following is not an access modifier?

- a) Protected
- ☒ b) Void
- c) Public
- d) Private

(x) Who invented Java Programming ?

- a) Guido van Rossum
- ☒ b) James Gosling
- c) Dennis Ritchie
- d) Bjarne Stroustrup

Section-B

Short Answer type Questions

2. Answer any **four** questions of the following :

5×4=20

S309/9/7

(5)

(Turn over)

- (a) What is a Constructor? How many types of constructors are used in Java?
- (b) What is a function Overloading? Explain it with example.
- (c) What are Applets in Java? Explain with proper example.
- (d) Write a program in Java to swap two numbers without using third variable.
- (e) What is the difference between System.out, System.err and System.in?
- (f) What are the advantages of Packages in Java?

Section-C

Long Answer type Questions

3. Answer any **two** questions of the following :

15×2=30

- (a) What is inheritance and encapsulation? Explain different types of inheritance with suitable diagram.

S309/9/7

(6)

Contd.

- (b) Explain Applet life cycle with suitable diagram and also describe
 - (i) Codebase,
 - (ii) Alt
 - (iii) Width
 - (iv) Code
- (c) Explain JDBC architecture in details.
- (d) Write a program in Java to find the factorial of a given number and also to check whether a number is prime or not.

———— x ————

S309/9/7

(7)

(P-500)